P-192 principal completions of the marked cyclic quotient

P-192 cyclic quotient, type-(1,r) polarization
Exactly two rational completion kernels for this polarization

Rational eigenspace completion -> standard product
Known composite map: diag(pi-1,pi-1)

Mixed eigenspace completion -> anti-isometry gluing
Quotient model unresolved; six explicit product controls supplied

Derived classification. Let V=E[r]^2 over the algebraic closure, with the alternating form from the product Weil pairing. Since p!=1 mod r, Frobenius is semisimple with two two-dimensional eigenspaces V_1 and V_p. The rational subgroup Gamma_x=<(G,Q)> lies in V_1. Each eigenspace is isotropic and the pairing between them is perfect: Frobenius multiplies Weil-pairing exponents by p, which rules out a nonzero pairing within either eigenspace. [pcc:unverified]

There are exactly two F_p-rational degree-r principal completions of the descended type-(1,r) polarization on A_x=E^2/Gamma_x, counted by their kernels. Indeed, a completion corresponds to a Frobenius-stable maximal isotropic W in V containing Gamma_x. Semisimplicity splits W into its two eigenspace parts. Either W=V_1, or W=Gamma_x plus the unique line in V_p orthogonal to Gamma_x. Equivalently, Gamma_x^perp/Gamma_x is two-dimensional with distinct eigenvalues 1 and p, so only its two eigenlines are Frobenius-stable. This concerns completions with pullback equal to the fixed descended polarization, not arbitrary polarizations or all isogenies of A_x. [pcc:unverified]

Explicit mixed kernel. Choose a geometric point T with pi(T)=pT and nontrivial e_r(G,T). In coordinates (G in first factor, G in second factor, T in first factor, T in second factor), the known line is (1,x,0,0). The mixed completion kernel is spanned by (1,x,0,0) and (0,0,-x,1). For x!=0 it is the graph of the anti-isometry psi_x(G)=xG, psi_x(T)=-x^(-1)T. The pairing multiplier is -1, and the action commutes with Frobenius. Thus it supplies a precise Frey-Kani gluing target as a group scheme. It does not supply a curve model, a quotient evaluator, or the missing action on T for an unknown x. [pcc:unverified]

The product route. W=V_1 is the kernel of diag(alpha,alpha), alpha=pi-1. Since bar(alpha)*alpha=r, the product polarization pulls back to r times itself. Hence this completion returns to E^2 with its product principal polarization. No unknown scalar is needed to describe this composite map. [pcc:unverified]

Six explicit mixed-kernel product controls. For u,v in O=Z[pi] with Norm(u)+Norm(v)=r, set M=[[u,v],[-bar(v),bar(u)]]. Then M^dagger M=rI and det(M)=r. If its rational and p-eigenvalue kernels equal the two mixed lines, it gives an explicit completion back to the standard product E^2. The implementation checks these identities and both formal kernel generators, plus actual evaluation on the rational pair (G,Q). This is a sufficient product-return certificate; it is not an exhaustive classification of reducible gluings or products with different elliptic factors. [pcc:unverified]

Integer controls. The exact identity r=a^2+b^2 holds with a=492336810215190858355684517 59 and b=62073717372636803831439154700. Taking u=a and v=+b or -b gives two mixed-kernel slopes x=-a/v mod r, satisfying x^2=-1. These are constructed control scalars, not assumptions about a supplied target. [pcc:unverified]

CM controls. Put N=t-2. The exact identity N=A^2+B^2 holds with A=106413589897786 and B=142420329313549. Choose n=A^2 or B^2, m=+AB or -AB, and u=pi-(n+1), v=m. Then Norm(u)=r-n(N-n), while m^2=n(N-n), so Norm(u)+Norm(v)=r. The four slopes x=n/m give four additional explicit mixed-kernel product completions. All six slopes are distinct. An exploratory factorization returned N itself as a prime factor; the independent checker uses only the sum-of-squares identities, so no primality certificate for N is required or claimed. [pcc:unverified]

Exact verification. Five public P-192 slopes from round50 are checked in the formal symplectic module, giving ten isotropic/Frobenius-stable subspace checks. Over F_11, an independent rank calculation checks every one of the twelve geometric quotient lines for each of eleven slopes: exactly two are stable in every case. All six product controls satisfy the exact CM polarization equation, both formal eigenspace kernel equations, and the actual rational point equations. Formal T coordinates are module coordinates, not an implementation of an extension-field point. [pcc:unverified]

Extension-field boundary. A nonzero T in the p-eigenspace is defined over F_(p^k) exactly when p^k=1 mod r. Direct modular checks exclude every 1<=k<=1000 for P-192. The exact multiplicative order is not computed here. This is a bound for explicit eigenpoint representation, not a lower bound on symbolic subgroup-scheme algorithms: a rational kernel can have non-rational generators. The known image G->Q does not by itself provide the second anti-isometry image T->-x^(-1)T. [pcc:unverified]

Literature. Achter and Howe, Split abelian surfaces over finite fields and reductions of genus-2 curves, section 5, reviews that a torsion anti-isometry has a maximal isotropic graph and that the multiplied product polarization descends to a principal polarization. Source: https://msp.org/ant/2017/11-1/ant-v11-n1-s.pdf , printed pages 50-51. That construction supplies the geometric framework; the explicit P-192 completion classification and control matrices above have their stated algebraic derivations and finite checks.

Next construction. Investigate a succinct symbolic description of the mixed kernel or an auxiliary isogeny method that constructs its quotient from G,Q without explicitly computing the unknown anti-isometry on the second eigenspace. Any claimed genus-2 target must also pass a reducibility check, including the six explicit product controls. The smaller CM class number could only help after this marked construction step has a concrete algorithm and cost.